

GRU (Gated Recurrent Unit)

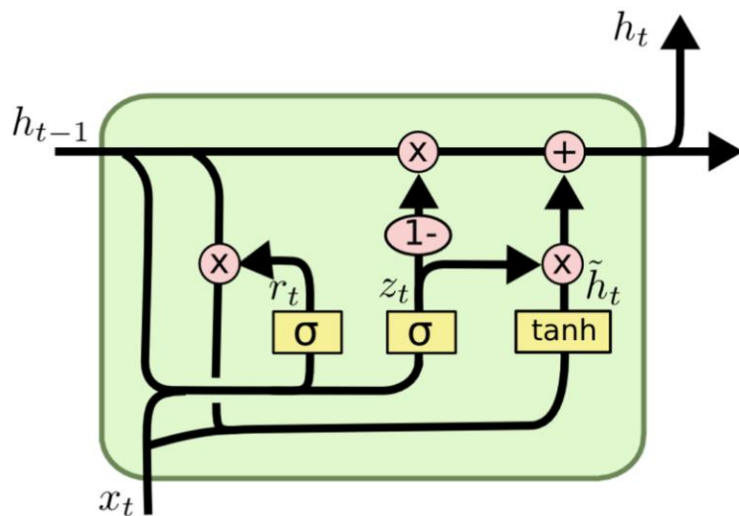
Fall 2025

School of IT Convergence

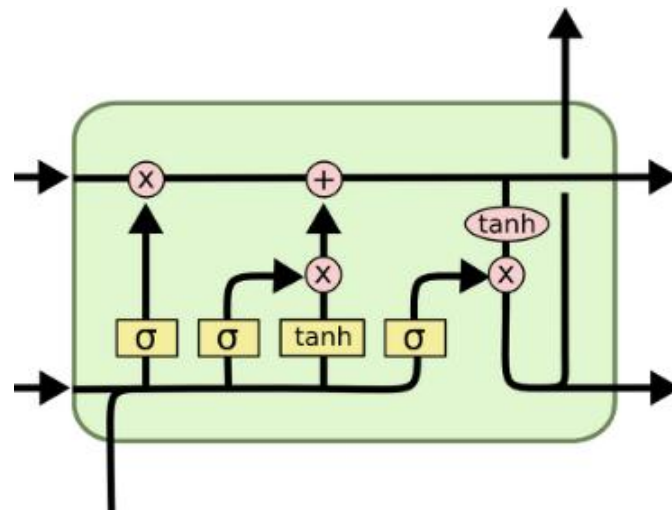
Prof. Dongsup Jin

GRU (Gated Recurrent Unit) (1)

- GRU (Gate Recurrent Unit)
 - Two gates to each neuron in the hidden layer
 - Reset gate
 - Update gate



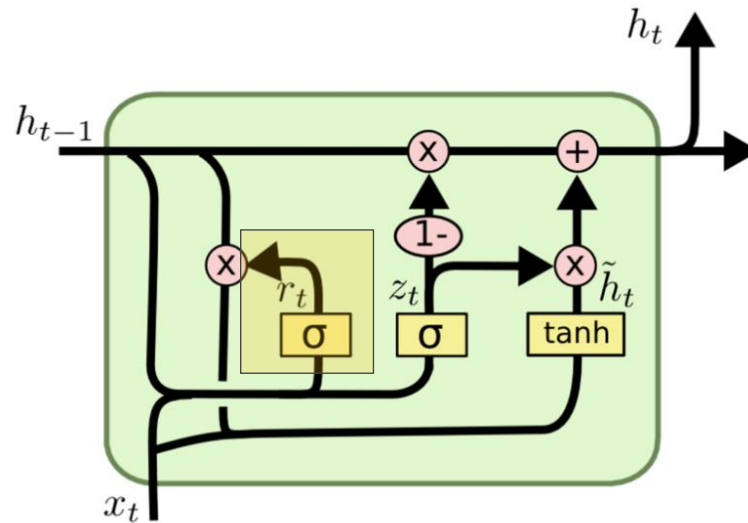
GRU hidden node



LSTM hidden node

GRU (Gated Recurrent Unit) (2)

- Reset gate
 - Reset the hidden state in the t-1
 - Use sigmoid function
 - Structure
 - Input: h_{t-1} and x_t
 - Output: r_t



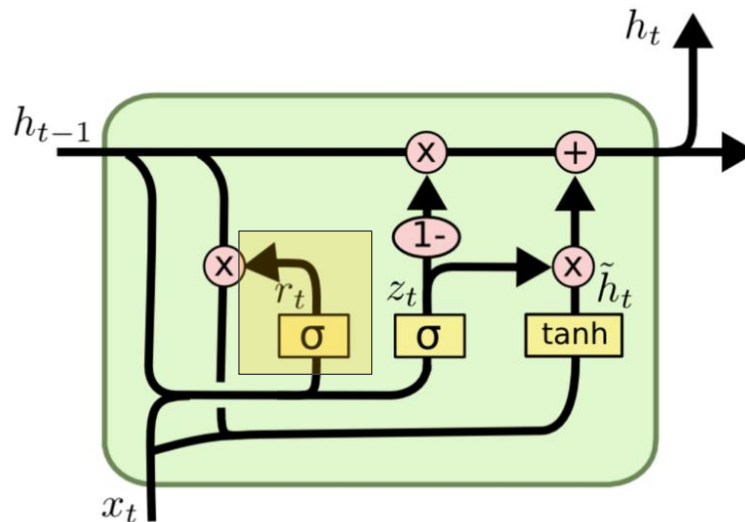
GRU hidden node

GRU (Gated Recurrent Unit) (3)

- Reset gate

- h_{t-1} : hidden vector in the t-1
- x_t : input vector
- W_r : **weight, bias omitted**
- σ : sigmoid function

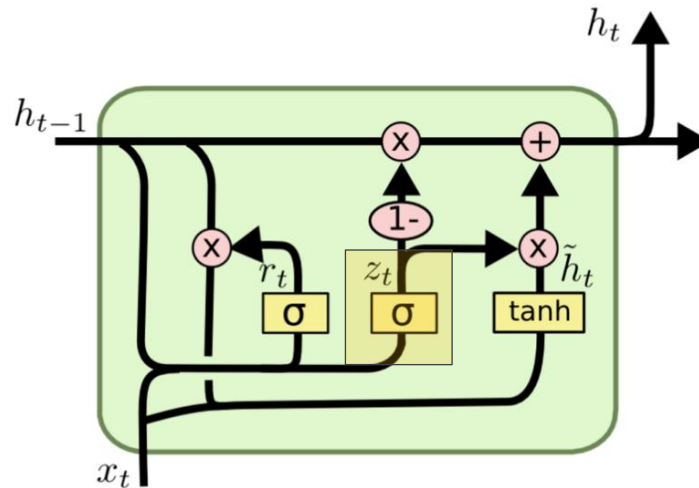
$$r_t = \sigma (W_r \cdot [x_t, h_{t-1}])$$



GRU hidden node

GRU (Gated Recurrent Unit) (4)

- Update gate
 - Determine the ratio of t-1 and t information
 - Be similar to the combination of forget gate and input gate in LSTM
 - Sigmoid function
 - Structure
 - Input: h_{t-1} and x_t
 - Output: z_t



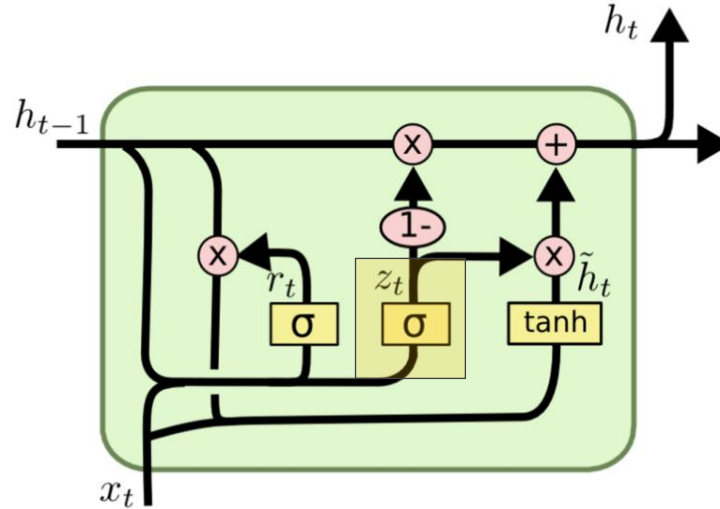
GRU hidden node

GRU (Gated Recurrent Unit) (5)

- Update gate

- h_{t-1} : hidden vector in the t-1
- x_t : input vector
- W_z : **weight, bias omitted**
- σ : sigmoid function

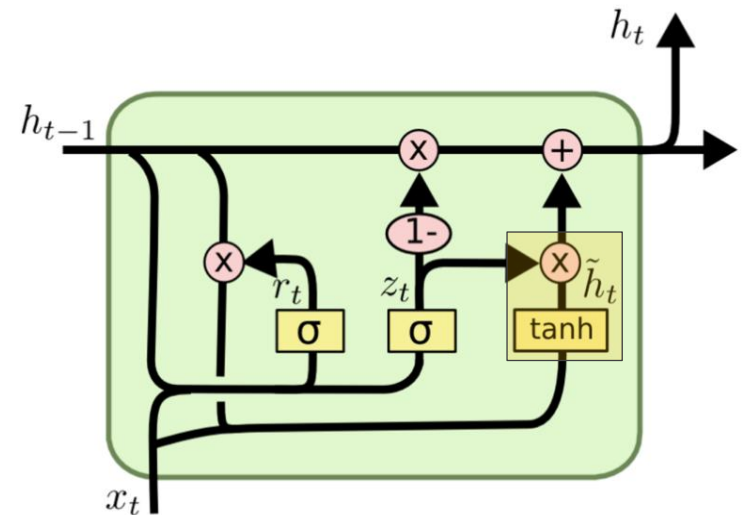
$$z_t = \sigma(W_z \cdot [x_t, h_{t-1}])$$



GRU hidden node

GRU (Gated Recurrent Unit) (6)

- Candidate
 - Calculate candidates in the t
 - Use the input vector and the result vector by multiplying the reset gate vector and the hidden vector in the t-1
 - Tangent hyperbolic function
 - Structure
 - Input: h_{t-1} , x_t , and r_t
 - Output: $\tilde{h}_t$

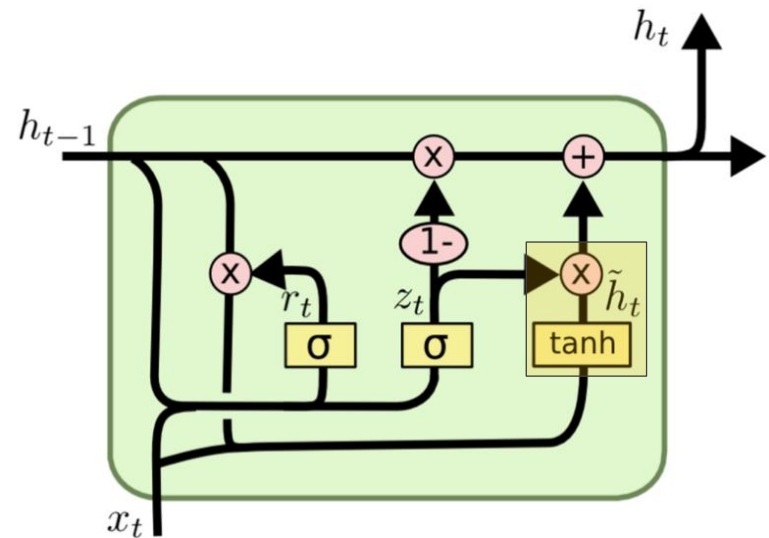


GRU hidden node

GRU (Gated Recurrent Unit) (7)

- Candidate
 - h_{t-1} : hidden vector in the t-1
 - x_t : input vector
 - r_t : reset gate result vector
 - W : **weight, bias omitted**
 - $\tanh$: tangent hyperbolic function

$$\tilde{h}_t = \tanh (W \cdot [x_t, r_t * h_{t-1}])$$

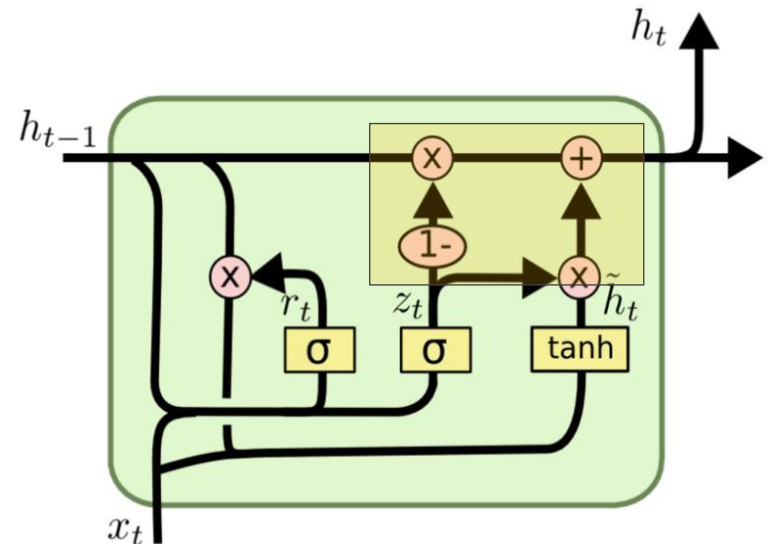


GRU hidden node

GRU (Gated Recurrent Unit) (8)

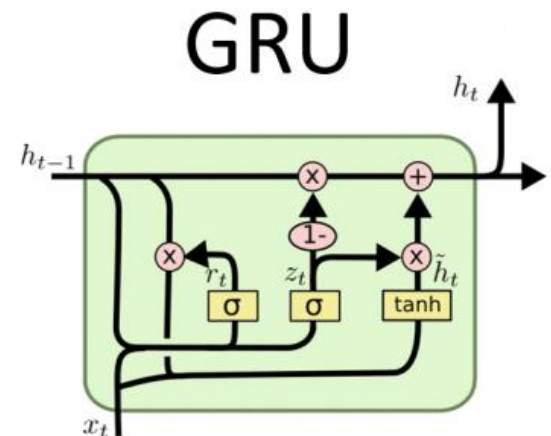
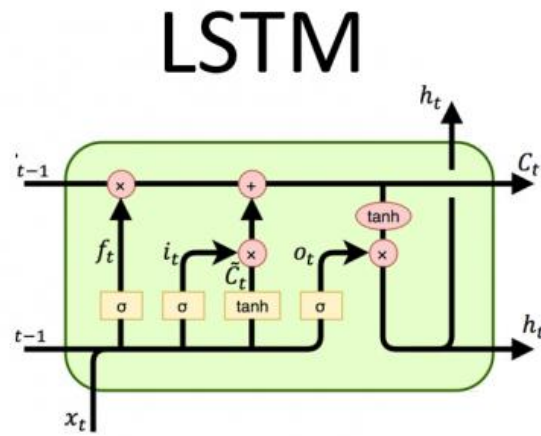
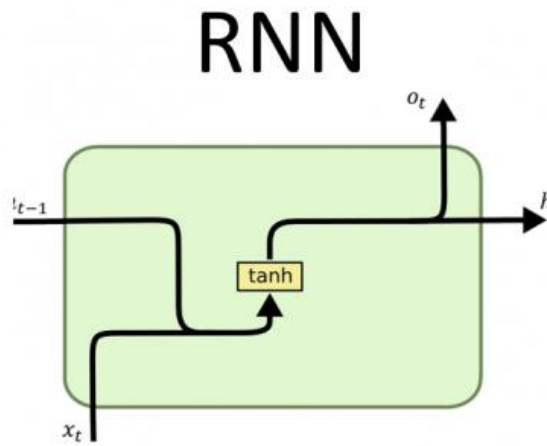
- Output
 - Calculate the hidden vector in the t by combining the updated gate result and candidate result
 - Structure
 - Input: h_{t-1} , $\tilde{h}_t$, and z_t
 - Output: h_t

$$h_t = (1 - z_t) * h_{t-1} + z_t * \tilde{h}_t$$



GRU hidden node

RNN vs LSTM vs GRU



Q & A
